

WORKBOOK DBMS ASSIGNMENT -1

Q1) Write SQL commands to create a database named SchoolDB. The program should: 1. Create a new database named SchoolDB. 2. Select the database for use. 3. Display confirmation by creating a simple table inside it.

```
CREATE DATABASE SchoolDB;

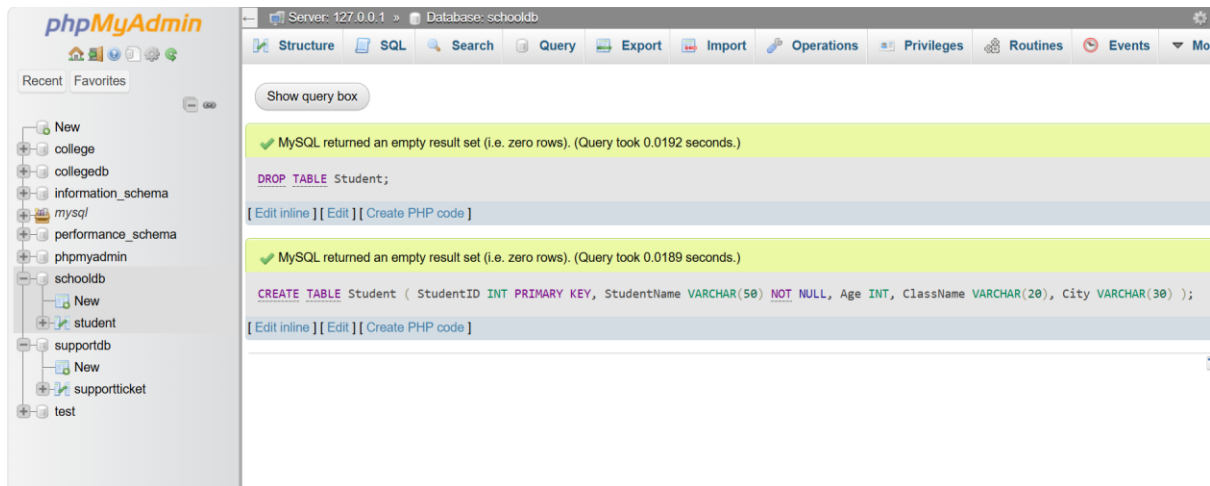
USE SchoolDB;

CREATE TABLE Student (
    StudentID INT PRIMARY KEY,
    StudentName VARCHAR(50),
    Age INT
);

SHOW TABLES;
```

Q2) Create a table named Student with the following columns: Column Name Data Type StudentID INT Constraint StudentName PRIMARY KEY VARCHAR(50) NOT NULL Age ClassName VARCHAR(30) INT VARCHAR(20) City The program should: 1. Create the table. 2. Define proper data types. 3. Apply primary key on StudentID.

```
Create Table Student (
StudentID INT PRIMARY KEY,
studentName varchar(50) NOT NULL,
Age INT,
ClassName VARCHAR(20),
City VARCHAR(30)
);
```



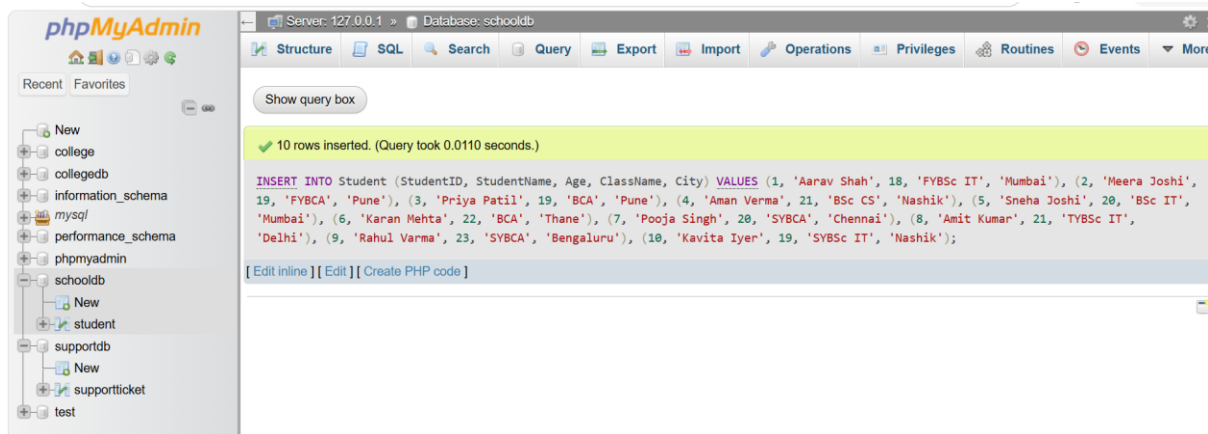
QUESTION 3

Q3) Insert at least 10 records into the Student table. The records should include: 1. Student ID 2. Student name 3. Age 4. Class name 5. City After inserting data, display all records using the SELECT command.

INSERT INTO Student (StudentID, StudentName, Age, ClassName, City)

VALUES

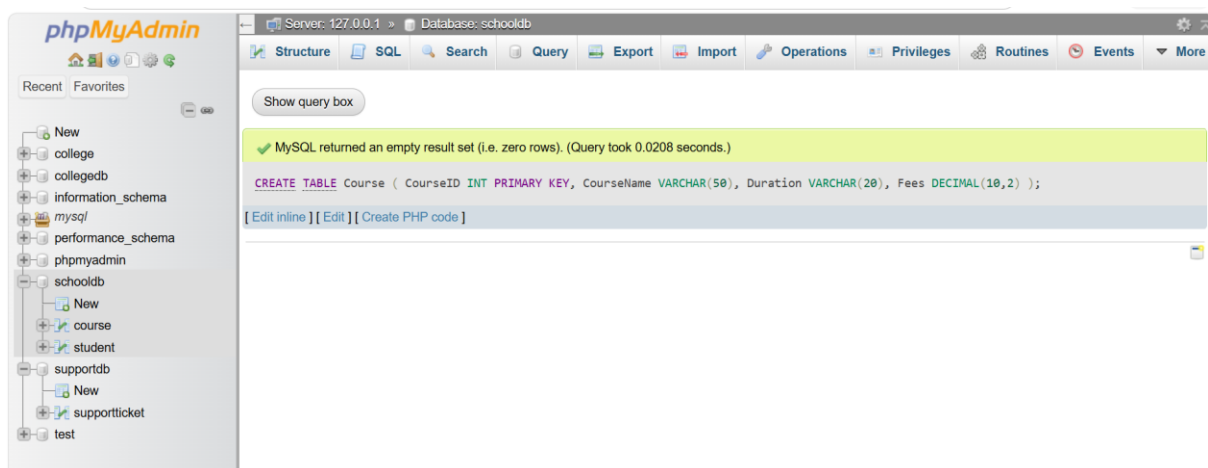
- (1, 'Aarav Shah', 18, 'FYBSc IT', 'Mumbai'),
- (2, 'Meera Joshi', 19, 'FYBCA', 'Pune'),
- (3, 'Priya Patil', 19, 'BCA', 'Pune'),
- (4, 'Aman Verma', 21, 'BSc CS', 'Nashik'),
- (5, 'Sneha Joshi', 20, 'BSc IT', 'Mumbai'),
- (6, 'Karan Mehta', 22, 'BCA', 'Thane'),
- (7, 'Pooja Singh', 20, 'SYBCA', 'Chennai'),
- (8, 'Amit Kumar', 21, 'TYBSc IT', 'Delhi'),
- (9, 'Rahul Varma', 23, 'SYBCA', 'Bengaluru'),
- (10, 'Kavita Iyer', 19, 'SYBSc IT', 'Nashik');



QUESTION 4

**Q4) Create a table named Course with the following fields: Column Name Data Type
CourseID CourseName INT VARCHAR(50) Duration Fees VARCHAR(20) DECIMAL(10,2) The program should: 1. Create the table. 2. Add a primary key on CourseID. 3. Insert at least 5 course records. 4. Display all course records.**

```
CREATE TABLE Course (
CourseID INT PRIMARY KEY,
CourseName VARCHAR(50),
Duration VARCHAR(20),
Fees DECIMAL(10,2)
);
```



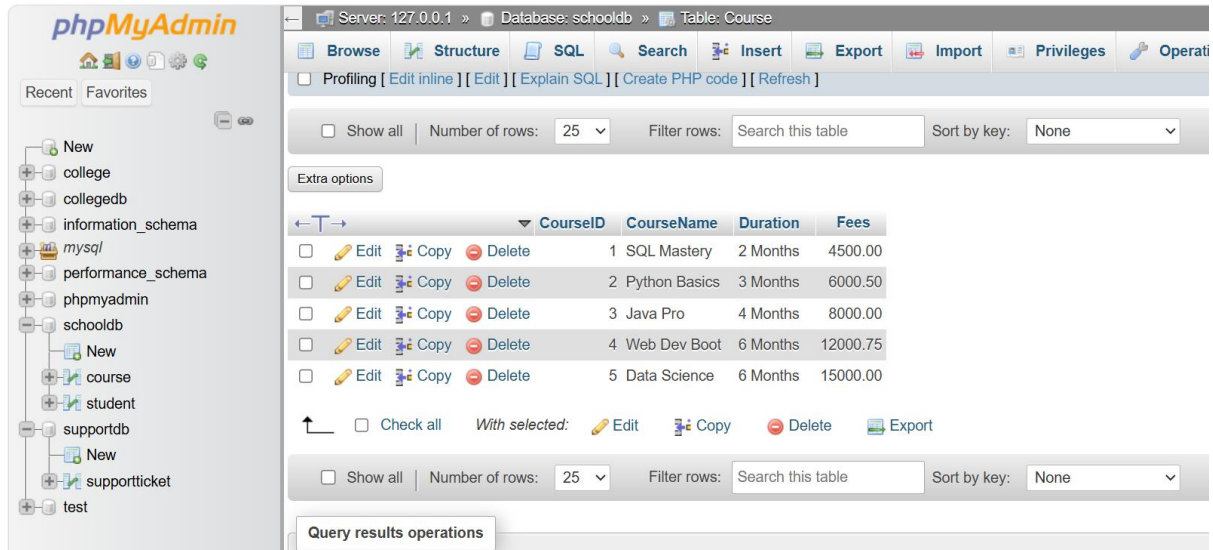
```
INSERT INTO Course (CourseID, CourseName, Duration, Fees)
VALUES
```

```
(1, 'SQL Mastery', '2 Months', 4500.00),
(2, 'Python Basics', '3 Months', 6000.50),
(3, 'Java Pro', '4 Months', 8000.00),
```

(4, 'Web Dev Boot', '6 Months', 12000.75),

(5, 'Data Science', '6 Months', 15000.00);

SELECT * FROM Course;



Server: 127.0.0.1 » Database: schooldb » Table: Course

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

		CourseID	CourseName	Duration	Fees
☐	Edit	1	SQL Mastery	2 Months	4500.00
☐	Edit	2	Python Basics	3 Months	6000.50
☐	Edit	3	Java Pro	4 Months	8000.00
☐	Edit	4	Web Dev Boot	6 Months	12000.75
☐	Edit	5	Data Science	6 Months	15000.00

Check all | With selected: Edit Copy Delete Export

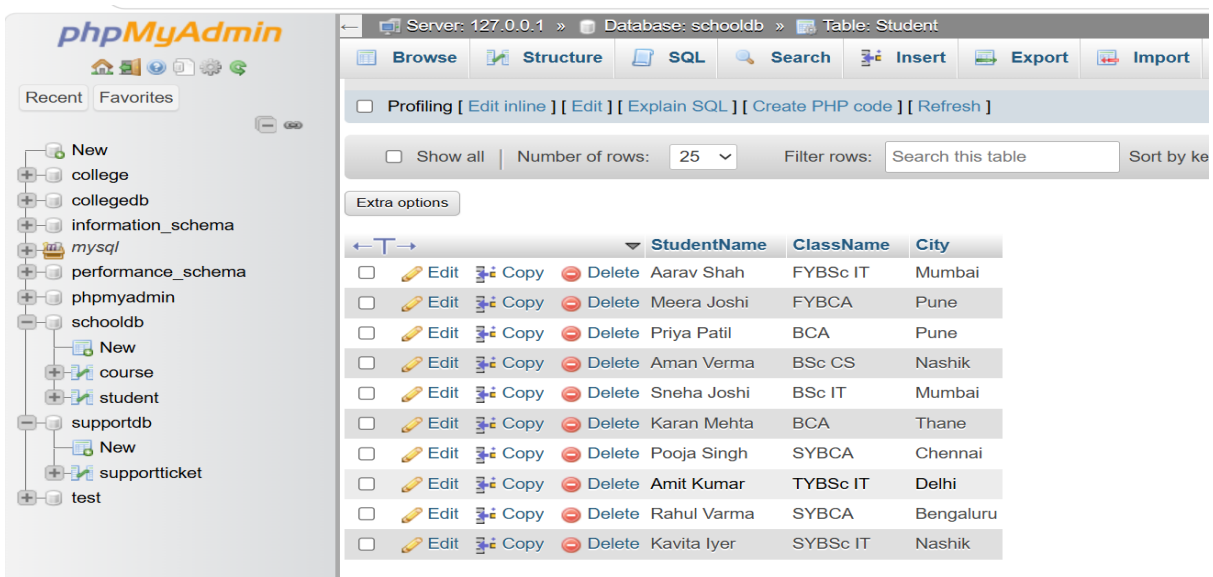
Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Q5) Write SQL queries to display only selected columns from the Student table. Display: 1. StudentName 2. ClassName 3. City Expected output should show only these three columns.

Select StudentName, ClassName, City

FROM Student;



Server: 127.0.0.1 » Database: schooldb » Table: Student

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

		StudentName	ClassName	City
☐	Edit	Aarav Shah	FYBSc IT	Mumbai
☐	Edit	Meera Joshi	FYBCA	Pune
☐	Edit	Priya Patil	BCA	Pune
☐	Edit	Aman Verma	BSc CS	Nashik
☐	Edit	Sneha Joshi	BSc IT	Mumbai
☐	Edit	Karan Mehta	BCA	Thane
☐	Edit	Pooja Singh	SYBCA	Chennai
☐	Edit	Amit Kumar	TYBSc IT	Delhi
☐	Edit	Rahul Varma	SYBCA	Bengaluru
☐	Edit	Kavita Iyer	SYBSc IT	Nashik

Q6) Write SQL commands to view the structure of the Student table. Use the suitable command according to the database environment: Database Command MySQL DESC Student; PostgreSQL tudent SQLite PRAGMA table_info(Student);

--MYSQL Environment

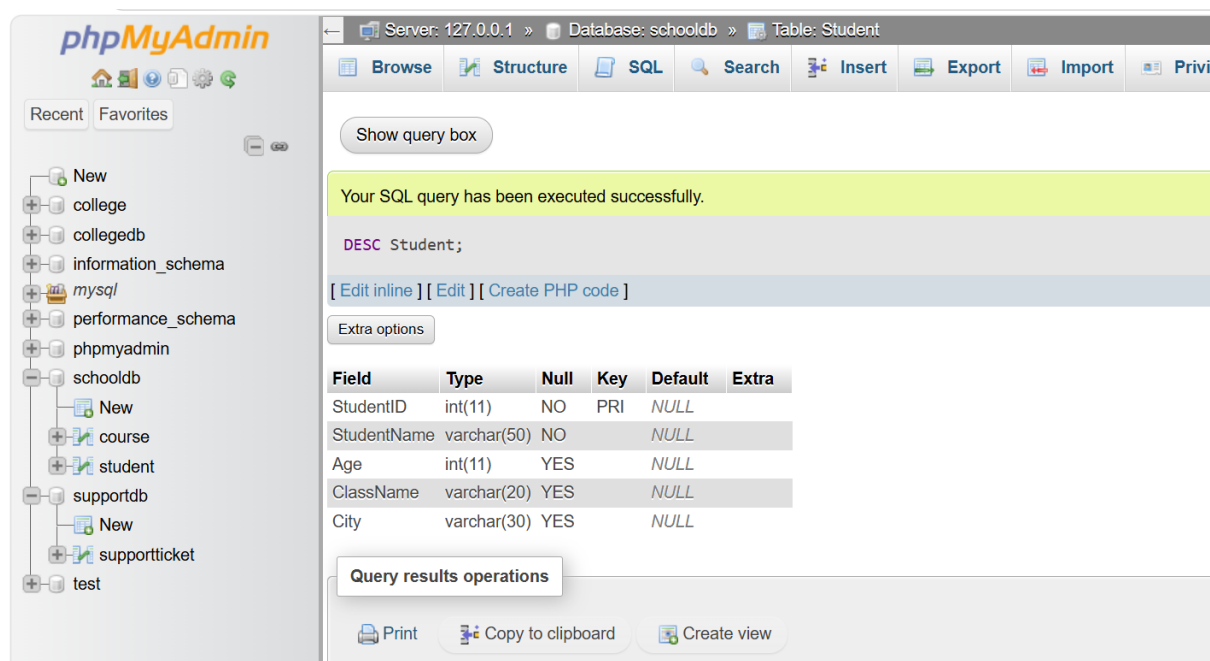
DESC Student;.....(only run this)

--postgreSQL Environment

\d Student

--SQLite Environment

PRAGMA table_info(Student);



Q7) Prepare a CSV file named students.csv and import it into the Student table. The CSV file should contain: StudentID StudentName 1 Aarav Shah Age Class Name City 18 2 Meera Joshi FYBSc IT Mumbai 19 3 Rohan Patil 20 FYBCA Pune SYBSc IT Nashik The task should include: 1. Creating a matching table. 2. Importing the CSV file. 3. Displaying imported records using SELECT.

LOAD DATA INFILE "C:/Users/wadek/Documents/Students.csv"

INTO TABLE Student

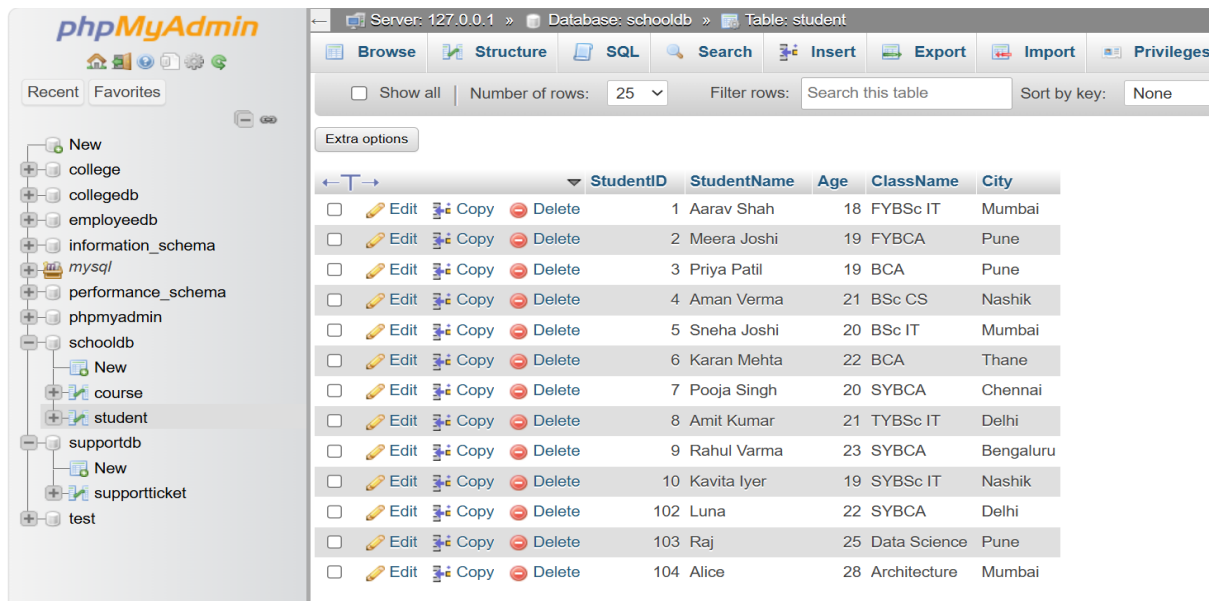
FIELDS TERMINATED BY ','

ENCLOSED BY '"'

LINES TERMINATED BY '\n'

IGNORE 1 ROWS;

SELECT * FROM Student;



Server: 127.0.0.1 » Database: schooldb » Table: student

Browse Structure SQL Search Insert Export Import Privileges

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	StudentID	StudentName	Age	ClassName	City
☐ Edit ☐ Copy ☐ Delete	1	Aarav Shah	18	FYBSc IT	Mumbai
☐ Edit ☐ Copy ☐ Delete	2	Meera Joshi	19	FYBCA	Pune
☐ Edit ☐ Copy ☐ Delete	3	Priya Patil	19	BCA	Pune
☐ Edit ☐ Copy ☐ Delete	4	Aman Verma	21	BSc CS	Nashik
☐ Edit ☐ Copy ☐ Delete	5	Sneha Joshi	20	BSc IT	Mumbai
☐ Edit ☐ Copy ☐ Delete	6	Karan Mehta	22	BCA	Thane
☐ Edit ☐ Copy ☐ Delete	7	Pooja Singh	20	SYBCA	Chennai
☐ Edit ☐ Copy ☐ Delete	8	Amit Kumar	21	TYBSc IT	Delhi
☐ Edit ☐ Copy ☐ Delete	9	Rahul Varma	23	SYBCA	Bengaluru
☐ Edit ☐ Copy ☐ Delete	10	Kavita Iyer	19	SYBSc IT	Nashik
☐ Edit ☐ Copy ☐ Delete	102	Luna	22	SYBCA	Delhi
☐ Edit ☐ Copy ☐ Delete	103	Raj	25	Data Science	Pune
☐ Edit ☐ Copy ☐ Delete	104	Alice	28	Architecture	Mumbai

Q8) Create a database named EmployeeDB and a table named Employee. The table should contain: Column Name Data Type EmployeeID INT EmployeeName VARCHAR(50) Department Salary VARCHAR(30) DECIMAL(10,2) Column Name Data Type City VARCHAR(30) The program should: 1. Create the database. 2. Create the Employee table. 3. Insert 8 employee records. 4. Display all employee records.

CREATE DATABASE EmployeeDB;

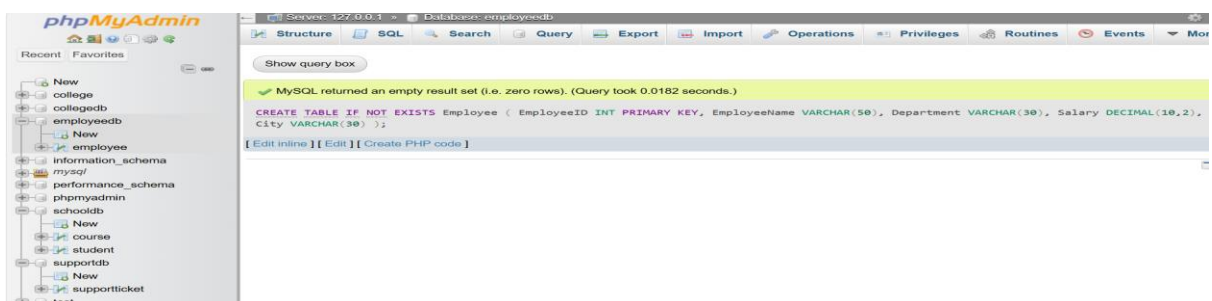
USE EmployeeDB;

CREATE TABLE Employee (

EmployeeID, INT PRIMARY KEY, EmployeeName VARCHAR(50),

Department VARCHAR(30), Salary DECIMAL(10,2), City VARCHAR(30)

);



Server: 127.0.0.1 » Database: employeedb

Structure SQL Search Query Export Import Operations Privileges Routines Events Mor

Show query box

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0182 seconds.)

```
CREATE TABLE IF NOT EXISTS Employee ( EmployeeID INT PRIMARY KEY, EmployeeName VARCHAR(50), Department VARCHAR(30), Salary DECIMAL(10,2), City VARCHAR(30) );
```

[Edit inline] [Edit] [Create PHP code]

INSERT INTO Employee

VALUES

(101, 'John Doe', 'IT', 75000.00, 'New York'),

```

(102, 'Jane Smith', 'HR', 65000.00, 'Chicago'),
(103, 'Sam Brown', 'Finance', 82000.00, 'Boston'),
(104, 'Lisa Ray', 'IT', 78000.00, 'Seattle'),
(105, 'Mark Lee', 'Sales', 60000.00, 'Austin'),
(106, 'Anna Kim', 'Marketing', 68000.00, 'Denver'),
(107, 'Paul Jay', 'Finance', 81880.00, 'Miami'),
(108, 'Emma Watson', 'HR', 66000.00, 'Dallas');

SELECT * FROM Employee;

```

The screenshot shows the phpMyAdmin interface for a database named 'employeeedb'. The 'Table: Employee' is selected, and the 'SQL' tab is active. The table structure is displayed with columns: EmployeeID, EmployeeName, Department, Salary, and City. The data is listed in a table with 8 rows, each with edit, copy, and delete icons. Below the table, there are options to check all, edit, copy, delete, and export. At the bottom, there are options to print, copy to clipboard, export, display chart, and create view.

EmployeeID	EmployeeName	Department	Salary	City
101	John Doe	IT	75000.00	New York
102	Jane Smith	HR	65000.00	Chicago
103	Sam Brown	Finance	82000.00	Boston
104	Lisa Ray	IT	78000.00	Seattle
105	Mark Lee	Sales	60000.00	Austin
106	Anna Kim	Marketing	68000.00	Denver
107	Paul Jay	Finance	81880.00	Miami
108	Emma Watson	HR	66000.00	Dallas

Q9) Write SQL queries to compare the difference between SELECT * and selecting specific columns. The task should include: 1. Query using SELECT * FROM Student; 2. Query using SELECT StudentName, City FROM Student; 3. Write two lines explaining the difference between both outputs.

--query 1: the wildcard approach

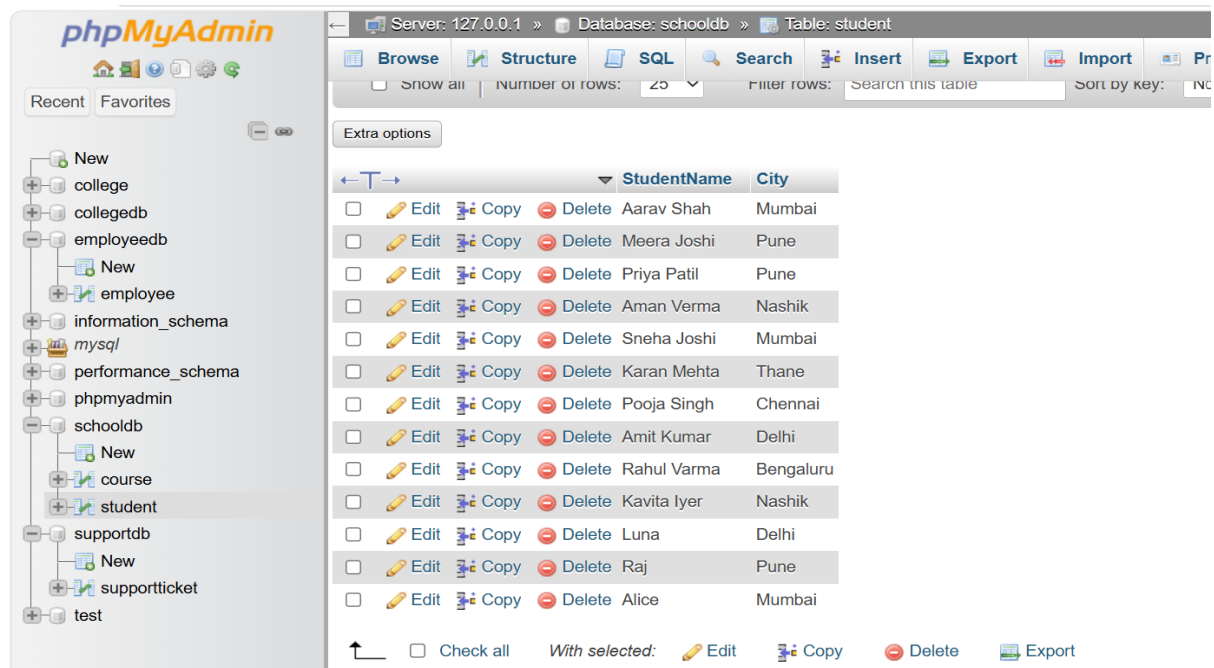
Select * FROM Student



--query2: the specific approach

Select StudentName, City

From student;



Q10) Create a complete SQL script for a mini college database. The script should include: 1. Creating database MiniCollegeDB. 2. Creating tables Student, Course, and Faculty. 3. Adding suitable columns and data types. 4. Inserting at least 5 records in each table. 5. Displaying all records from each table. 6. Viewing the structure of each table.

CREATE DATABASE MiniCollegeDB;

USE MiniCollegeDB;

```
CREATE TABLE Student (SID INT PRIMARY KEY, SName VARCHAR(50));
```

```
CREATE TABLE Course (CID INT PRIMARY KEY, CName VARCHAR(50));
```

```
CREATE TABLE Faculty (FID INT PRIMARY KEY, FName VARCHAR(50));
```

```
INSERT INTO Student
```

```
VALUES (1, 'Ali'), (2, 'Ben'), (3, 'Cam'), (4, 'Dan'), (5, 'Eve');
```

```
INSERT INTO Course
```

```
VALUES (10, 'Math'), (11, 'Sci'), (12, 'Art'), (13, 'Lit'), (14, 'CS');
```

```
INSERT INTO Faculty
```

```
VALUES (100, 'Dr.X'), (101, 'Dr.Y'), (102, 'Dr.Z'), (103, 'Dr.A'), (104, 'Dr.B');
```

```
SELECT * FROM Student; SELECT FROM Course; SELECT FROM Faculty;
```

```
DESC Student; DESC Course; DESC Faculty;
```

The image displays two screenshots of the phpMyAdmin web interface, version 4.7.0.1, running on a MySQL server (127.0.0.1). The left sidebar shows a tree view of databases, including 'college', 'collegedb', 'employeedb', 'employee', 'information_schema', 'mysql', 'performance_schema', 'phpmyadmin', 'schooldb', 'course', 'student', 'supportdb', 'supportticket', and 'test'.

The top screenshot shows the 'Student' table selected. The table has two columns: 'SID' and 'SName'. The data is as follows:

SID	SName
1	Ali
2	Ben
3	Cam
4	Dan
5	Eve

The bottom screenshot shows the 'Course' table selected. The table has two columns: 'CID' and 'CName'. The data is as follows:

CID	CName
10	Math
11	Science
12	Art
13	Literature
14	Computer Science

phpMyAdmin

Recent Favorites

- New
- college
- collegedb
- employeeedb
 - New
 - employee
- information_schema
- mysql
- performance_schema
- phpmyadmin
- schoolodb
 - New
 - course
 - student
- supportodb
 - New
 - supportticket
- test

Server: 127.0.0.1

Databases SQL Status User accounts Export

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

		FID	FName
☐	Edit Copy Delete	100	Dr. X
☐	Edit Copy Delete	101	Dr. Y
☐	Edit Copy Delete	102	Dr. Z
☐	Edit Copy Delete	103	Dr. A
☐	Edit Copy Delete	104	Dr. B

☐ Check all | With selected: Edit Copy Delete

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

phpMyAdmin

Recent Favorites

- New
- college
- collegedb
- employeeedb
 - New
 - employee
- information_schema
- mysql
- performance_schema
- phpmyadmin
- schoolodb
 - New
 - course
 - student
- supportodb
 - New
 - supportticket
- test

Server: 127.0.0.1

Databases SQL Status User accounts Export Import Set

Field	Type	Null	Key	Default	Extra
SID	int(11)	NO	PRI	NULL	
SName	varchar(50)	YES		NULL	

Query results operations

Print Copy to clipboard Create view

Your SQL query has been executed successfully.

DESC Course;

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
CID	int(11)	NO	PRI	NULL	
CName	varchar(50)	YES		NULL	

Query results operations

Your SQL query has been executed successfully.

DESC Faculty;

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
FID	int(11)	NO	PRI	NULL	
FName	varchar(50)	YES		NULL	